CP255 - W2_S2 Python Basics Answer Key

Your Turn: Calculate Seconds in a Day

- How many seconds are in one day?
- Hint: 60 seconds per minute, 60 minutes per hour, 24 hours per day

Answer:

```python
# TODO: Calculate the number of seconds in a day
# Write your expression below:

60 * 60 * 24
```

Your Turn: Create Your Own Variables
Create variables that references:

1. Your favorite number
2. The current year
3. How old you will be in 10 years

Answer: (variables can be set to different values, this is just an example)

```python
# TODO: Create your variables below

favorite_number = 22
current_year = 2026
my_age = 23

# Calculate age in 10 years
age_in_10_years = my_age + 10

age_in_10_years
```

Your Turn: BART Station Analysis

Now let's put it all together! Complete the exercises below.

Answer:

```python
# Exercise 1: Create variables for Rockridge station
# Fill in the values (you can look them up or estimate)

rockridge_name = "Rockridge Station"  # Station name as a string
rockridge_entries = 3200  # Daily entries as an integer (try 3200)
rockridge_lat = 37.8442  # Latitude as a float (try 37.8442)
rockridge_has_parking = True  # Does it have parking? (True/False)

# Print each variable with its type
print("Name:", rockridge_name, "- Type:", type(rockridge_name))
# Add more print statements...
print("Daily Station Entries:", rockridge_entries, "- Type:", type(rockridge_entries))
print("Station Latitude:", rockridge_lat, "- Type:", type(rockridge_lat))
print("Does station have parking:", rockridge_has_parking, "- Type:",
type(rockridge_has_parking))
```

```python
# Exercise 2: Calculate the difference in ridership

berkeley_entries = 4521
rockridge_entries = 3200

# Calculate how many more entries Berkeley has
difference = berkeley_entries - rockridge_entries
print("Berkeley has", difference, "more daily entries")

# Calculate Berkeley's entries as a percentage of Rockridge's
percentage = (berkeley_entries / rockridge_entries) * 100
print("Berkeley has", percentage, "% of Rockridge's ridership")
```

```python
# Exercise 3: String methods

station = "  macarthur BART station  "
```

```
# 1. Remove the extra spaces from both ends (hint: .strip())
cleaned = station.strip()

# 2. Convert to title case
titled = cleaned.title()

# 3. Replace "Bart" with "Transit"
replaced = titled.replace("Bart", "Transit")

print("Cleaned:", cleaned)
print("Titled:", titled)
print("Replaced:", replaced)
```